Emergence or Artifact? A Structure-Provenance Framework for Synthetic Organizations

Abstract

Large language model (LLM) agent simulations offer a setting for investigating how coordination, task specialization, and organizational networks develop. Yet persistent specialization or communication centrality does not reveal whether these patterns arise from members' histories and interactions, prescribed roles, unequal access to tasks, or incidental features of the simulation. We introduce structure-provenance analysis, an open framework for assessing the persistence of organizational differences and testing competing explanations for their origins. It combines comparisons with null models, robustness tests using changes to features with no assigned organizational significance, and controlled interventions on agents' roles, histories, and interactions. We evaluate its ability to distinguish known mechanisms in rule-based agents and show that its measures detect persistent member differences in interaction data from insect colonies, workplaces, and online teams.

Applications to LLM-agent organizations show why these distinctions matter. Initially equivalent agents completed interdependent tasks without detectable stable task specialization. Prescribing specialties produced persistent differentiation but reduced collective task completion under the tested conditions. An apparently stable communication hub disappeared under balanced relabeling, revealing a model preference for an arbitrary agent label rather than a particular member. These findings distinguish coordination from a stable division of labor and show how apparent network structure can reflect incidental cues.

Our open repository provides the measures, null models, complete simulation records, executable experiments, a one-command analysis tool, and a two-minute guided demonstration. Together, these resources provide a reproducible basis for investigating how organizational differentiation develops and for distinguishing emergent from built-in structure.

1. The problem of attributing organizational structure

Simulations with interacting large language model (LLM) agents increasingly examine network formation, task specialization, and authority stratification (Takata et al. 2024, Papachristou and Yuan 2025, Ji et al. 2026). They allow researchers to observe interaction histories and manipulate agents' instructions, available information, and experience. Yet observing a persistent division of labor or a central actor in a communication network does not identify the process that produced it. Such patterns may reflect agents' behavior and interactions, prescribed roles, or incidental features of the simulation design, such as labels, ordering, or routing.

This distinction matters for organizational theory. The division of labor involves both the decomposition of work and its allocation among members, while coordination concerns how their contributions are integrated (Puranam et al. 2014). Agents may therefore complete interdependent tasks without developing stable specializations. Likewise, the repeated concentration of communication around one member establishes a pattern of network centrality, not necessarily the development of authority. In both cases, the organizational outcome and its explanation require different evidence.

We develop structure-provenance analysis to assess organizational differentiation and investigate its origins. The framework compares persistent differences in activity, task specialization, and communication patterns with dimension-appropriate null models; examines whether patterns withstand changes to features with no assigned organizational significance; and tests competing explanations using controlled interventions on roles, histories, and interactions. We evaluate its diagnostics in rule-based systems with known generating mechanisms (Theraulaz et al. 1998), apply them to interaction records from insect colonies (Mersch et al. 2013, Gernat et al. 2018, Alciatore et al. 2021), workplaces (Génois et al. 2015, Génois and Barrat 2018), and online teams (Cui et al. 2017), and then use them to reassess LLM-agent organizations.

2. Structure-provenance analysis

The framework addresses three questions, each requiring different evidence.

- **Is there structure?** We assess persistent differentiation across four dimensions: members' levels of activity, task specialization, communication centrality, and the stability of interaction partners. The first three are assessed using repeatability, which captures the persistence of between-member differences across observation windows (Nakagawa and Schielzeth 2010); partner stability is assessed through overlap in interaction partners across windows. Each measure is compared with an appropriate null model: permutations of member identities within windows for repeatability, and random partner sets for partner stability. Activity is adjusted for opportunities to perform work, although unequal task exposure remains a limitation. Exposure-preserving and dependence-aware null models are planned for Stage 2 (D1).
- **Is it robust?** We test whether observed differences persist when features without assigned organizational significance are varied, including agent labels and the order in which tasks and colleagues are presented.
- **Where did it come from?** We use controlled interventions to distinguish competing explanations for observed differentiation. Persistent differences may arise from prescribed roles, unequal access to tasks or information, agents' responses to incidental cues, or differences acquired through experience and interaction. These sources may coexist. Interventions that reassign roles, reset agents' histories, restrict communication, or vary labels help identify their contributions, although an intervention can change more than one factor: restricting communication, for example, also changes which tasks agents see. Observational records alone cannot establish these origins.

The framework produces quantitative diagnostics rather than a categorical judgment of emergence. For the LLM comparisons in Section 3, primary outcomes and decision rules were specified in a written analysis plan before the new replication results were examined; analyses restricted to post-plan runs support the same conclusions. The anonymized repository (anonymous.4open.science/r/synthetic-org-benchmark-45E2) contains the analysis tools, experimental controls, complete records of the reported simulations, and a two-minute guided demonstration requiring no API access.

Contribution. Existing methodological work addresses experimental design, reproducibility, and validation in LLM-based simulations (Hu and Shen 2026, Tranchero et al. 2026). We address a distinct inferential problem: how to move from observing persistent organizational

differentiation to evidence about the processes that generate it. Structure-provenance analysis integrates persistence measures, null-model comparisons, robustness tests, and controlled interventions, evaluated against systems with known generating mechanisms. It thereby provides a common framework for assessing organizational structure across synthetic and empirical settings, and for investigating its origins where interventions are possible.

3. Evidence

We evaluate the framework on empirical interaction data, against known generating mechanisms, and in LLM-agent simulations. Full results are reported in the technical appendix.

A. Measurement across empirical groups. The persistence measures detect stable member-level differences across biologically and socially distinct groups, each compared with its own null model (Figure 1A); because the groups are scored on different behaviors, values are compared within groups, not across them. These include genetically identical ants (0.68 against a baseline of 0.13; Alciatore et al. 2021), employees in two workplaces (0.34 and 0.28 against 0.16 and 0.14; Génois et al. 2015, Génois and Barrat 2018), and strangers matched into online game teams (0.51 against 0.15; Cui et al. 2017). Preliminary analyses of additional ant and honeybee colonies show similar patterns (Mersch et al. 2013, Gernat et al. 2018). These applications establish the measures' usefulness across empirical settings, but the observational data alone cannot identify the processes producing the differences. In an independently developed LLM-agent simulation with self-written memory (Takata et al. 2024), the measures detect persistent differences that fall to chance when every agent hears every other.

B. Validation against known generating mechanisms. We evaluate whether the framework distinguishes prescribed from acquired specialization using rule-based agents implementing a response-threshold model of division of labor (Theraulaz et al. 1998). Across 20 runs per condition, persistent task specialization is not detected when agents have no source of stable differences but is detected in every run when task preferences are either prescribed or acquired through experience (Figure 1B). Resetting acquired response thresholds midway through a run reduces the continuity of specialization to chance, whereas prescribed roles remain unaffected (difference in continuity: 0.19, 95% interval 0.10–0.28). Restricting communication reduces specialization by 0.09 (0.06–0.13) without removing it, although the accompanying change in task opportunities prevents isolating the effect of interaction. Finally, rule-based agents given a built-in preference for the label "A1" form a persistent hub under fixed labels. Our initial relabeling left it intact, because one agent is never anyone's "A1;" balanced relabeling removed it in all 20 runs. The test thus exposed a flaw in our own perturbation and shows why a perturbation must be checked to remove the suspected source.



Figure 1. Markers are runs or sites, filled when above their own shuffled baseline (95th percentile of a label-permutation null) and hollow when not. Grey bars run to each row's median baseline; black marks show the median with a bootstrap 95% interval; the right column counts runs above baseline. (A) Empirical groups. (B) Rule-based agents with known sources of differentiation; for wiped histories, attribution rests on continuity across the wipe, which returns to chance (3B), while specialization can re-form afterwards. (C) Plan-specified comparisons of LLM organizations. Rows record different behaviors, so each is read against its own baseline.

C. Coordination without stable specialization. In the LLM simulations, nine initially equivalent agents completed interdependent tasks requiring communication between members. Yet neither of the two models exhibited detectable persistent task specialization in the 20 plan-specified runs (0/20 above the prespecified baseline; 1/20 under an alternative calibration). By contrast, prescribing task specialties produced persistent differentiation in every run (20/20; Figure 1C), but reduced task completion by 30–69% under the tested conditions. Because the intervention also constrained task allocation, this performance difference cannot be attributed to specialization alone. Moreover, because agents' capabilities were identical regardless of role, prescribed specialties imposed the constraint of specialization without the task-specific efficiency that usually accompanies it. The findings demonstrate that coordination can occur without a stable division of labor and that imposing differentiated roles need not improve collective performance.

D. Communication centrality and arbitrary labels. Under fixed agent names, gpt-4.1-mini agents formed a persistent communication hub in all ten plan-specified runs (Figure 1C), usually around the agent labeled "A1." Yet the apparent centrality disappeared when labels were balanced across agents. The preference for the label itself remained: 88 of 90 agents directed most of their messages to whichever colleague they encountered as "A1." Thus, the concentration of communication reflected systematic responses to an arbitrary identifier rather than the emergence of a consistently central member. All four plan-specified contrasts in C and D remain significant after Holm correction ($p < 0.001$); paired comparisons yield $p < 0.01$. Together, C and D show that two conventional organizational outcomes, coordination and centrality, can arise without the generating processes usually inferred from them.

4. Stage 2: Three deliverables

D1. The analysis package. We will finalize the four outcome measures and extend the null models to account for unequal task opportunities, temporal dependence, and label-related preferences in partner choice. The validated balanced relabeling procedure will replace the original. The package will include documented data formats, known-answer tests, a worked example, and a one-command report. Success requires that an independent researcher can reproduce the documented results and apply the diagnostics to a compatible interaction record.

D2. A controlled test of endogenous differentiation. We will investigate whether initially equivalent LLM agents develop persistent task specialization through their own experience and feedback, and whether colleagues' outcomes add to it. A factorial experiment varies persistent memory, outcome feedback, and access to colleagues' outcomes while holding agent capabilities, instructions, and task opportunities constant. Work passing, which changes task opportunities, will be examined separately. A further condition introduces task-specific efficiency gains from repetition and a cost of switching tasks. A condition with initially unknown comparative advantages will distinguish the development of specialization from adaptation to pre-existing differences.

Each condition will be run a preregistered number of times on a current proprietary and a current open-weights model; conditions, outcome measures, decision rules, and model versions will be preregistered. The Stage 1 comparisons (identical versus assigned-specialty agents, fixed versus balanced labels) will be repeated as reference tests and reported by model. Success consists of identifying which combinations of memory, feedback, and information about colleagues produce

detectable persistent specialization, including informative null results and model-specific differences, and whether replicate runs from identical starting conditions converge on the same configuration or on alternatives.

D3. Independent validation and reporting standards. We will apply the framework to at least two independently developed simulations, including one examined in Stage 1, and report how its diagnostics change their interpretation. Our team has experimental access to honeybee colonies. Replicated cohorts of newly emerged, same-age workers, with controlled variation in task opportunities and social experience and individual tracking, will test whether the diagnostics attribute differentiation correctly when its source is known in a biological group, not only in rule-based agents. We will also examine sensitivity to observation-window length, unequal task opportunities, and group size, and develop a one-page reporting standard for studies of synthetic organizations. Success requires demonstrating which inferences transfer beyond our environment and identifying where available data are insufficient to establish the origins of observed differences.

Stage 1's LLM experiments cost about $23; Stage 2 needs a few hundred dollars in model usage and existing laboratory resources.

5. Risks and safeguards

Exchangeability and temporal dependence. The permutation-based null models assume that members are exchangeable within observation windows. Adjusting activity for task opportunities mitigates, but does not eliminate, differences in exposure. D1 addresses both with extended null models.

Model- and task-specificity. Stage 1 establishes how two specified LLM models behaved in a particular organizational environment, not a general limit on LLM capabilities. Other models or task environments may produce stable specialization without prescribed roles or show less sensitivity to arbitrary labels. Such variation would itself be informative, because the framework identifies the conditions under which differentiation develops rather than assuming a universal outcome. Proprietary models may also be retired; the open-weights model in D2 keeps every result reproducible.

Reproducibility and data use. Model versions, settings, responses, and interaction histories are recorded. A clean-environment check reproduced the synthetic and stored LLM results from the public repository; the empirical results require the original data deposits. External data are used under their respective licenses and retain published pseudonymous identifiers. No new human-subjects data are collected.

6. From observed structure to its origins

Synthetic organizations allow controlled tests of the origins of organizational differences. Stage 1 shows that interdependent work can be coordinated without stable task specialization and that persistent communication centrality can reflect arbitrary agent labels. Stage 2 investigates when memory, feedback, and information about other members support the development of durable specialization. If it succeeds, claims about emergent roles and hierarchies in synthetic organizations will rest on evidence about their origins, and organization theory will gain an experimental setting for studying how initially equivalent members develop durable roles.

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